

Role of Cyclone “Sidr” on Chlorophyll distribution in Bay of Bengal

Abstract

From the prehistoric time scale the coast of Bangladesh had been hit with a substantial amount of cyclone, which has a diverse effect on the properties of coastal water as well as in the waters of Bay of Bengal. In this study, impact of Cyclone Sidr on the Bay of Bengal water and its surface chlorophyll concentration has been studied along with Bangladesh during October to December 2007 using Aqua-MODIS Ocean Color Web (OCB) with a resolution of 4 km. The total chlorophyll concentration found to be increasing with the effect of Sidr. The data has been correlated with MODIS and windspeed from Quikscat scatterometer respectively. There has been observation of 2-3°C decrease in SST with movement of cyclone. Whereas K_d has a substantial increase of 2-5 m^{-1} . The wind speed has peaked up to 140 knots in the cyclonic zone. Our results suggest that cyclone disturbed chlorophyll abundance in a consistent way and induce ordered and predictable changes in sea surface temperature as well as the diffusive attenuation coefficient rate.

Keywords: Cyclone Sidr, chlorophyll concentration, The diffuse attenuation coefficient (K_d), SST, wind speed.

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1. INTRODUCTION

The Super Cyclonic Storm named Sidr (TC06B-2007) originated from a depression at southsouthwest of Port Blair (Andaman and Nicobar Islands, India) in the Bay of Bengal (BoB) on November 11, 2007 (see Fig. 1). It struck the coast on November 15, 2007 in Bangladesh, which was one of the worst affected countries in the region.(Tummala, 2009)

The BoB is different from its western counterpart, the Arabian Sea (AS) because of higher SST, lower surface chlorophyll-a concentration (Chl-a) and salinity, and more cloud cover. Past studies on the impact of cyclone related events on primary productivity in the BoB region have shown that there is a significant increase in Chl-a concentrations following a cyclone. In addition, significant decreases (up to 6 °C) in SST were observed during October 1999 super cyclone event.(Tummala, 2009)

The BOB is a semi-enclosed tropical basin in the northern Indian Ocean. Due to the annually reversing monsoons, it undergoes seasonal fluctuations in circulation and weather. The study of the physical pushing and its effects has raised questions. The fact that the Bay receives a lot of freshwater from rivers and rains may be the cause of this variation. Cyclones frequently form in the Bay of Bengal during the southwest and northeast monsoon seasons, strike the shore, and completely destroy coastal resources. In the BOB, tropical cyclones (TCs) often form in the western and central regions of the basin during the pre-monsoon (April to May) and post-monsoon (October to December) (Sengupta et al,2008). Nearly 5% of TCs that happen each year throughout the world are accounts of BOB (Neetu et al. 2012).

Large-scale systems or windstorms known as tropical cyclones revolve around a low-pressure center over a vast region of warm tropical waters. These have an impact on marine phytoplankton productivity and abundance. This also has an impact on the oceanographic characteristics, both physical and biological. These cyclonic disturbances also show strong wind speed and heavy rainfall. In terms of biological consequences, the cyclone wind field causes local mixing which results in the injection of nutrients into the upper layer of the ocean and triggering phytoplankton blooms. The upper ocean response to a moving cyclone has been studied by many researchers. Previous studies over the BoB have reported remarkable change in Chlorophyll-a (Chl-a), sea surface temperature (SST) and Diffuse Attenuation Coefficient (KD) predominantly due to cyclones, depending on their path, strength, location and time of occurrence (Lotliker et al. 2014; Baliarsingh et al. 2015; Sarangi et al. 2015; Chacko 2017; Mandal et al. 2018; Bhowmick et al. 2020; Ganguly et al. 2021). The infusion of nutrients into the euphotic zone and the availability of adequate light conditions for photosynthesis at the surface as a result of cyclone-induced upwelling and enhanced vertical mixing support the condition of increased Chlorophyll-a concentration. Understanding ocean mixing and biogeochemical processes depends on research on phytoplankton blooms brought on by tropical storms. Strong cyclonic winds frequently cause a drop in sea surface temperature (SST), which has an impact on both physical and biological properties (Mandal et al.2018).

Due to the severe weather during the cyclone times, direct readings from ships are not feasible. The best method for analyzing oceanic fluctuations during these kinds of episodic events is satellite-based observation. In order to research the phenomenon of cyclones, satellite remote sensing gives data on a variety of factors, including sea surface temperature, wind speed, and ocean color datasets, such as chlorophyll-a concentration and Diffuse Attenuation Coefficient (KD). Satellites that measure ocean color have revolutionized our understanding of how tropical cyclones affect the distribution of chlorophyll, phytoplankton bloom, and other marine physical processes. Several sensors can currently be used to find cyclonic activity.

So, in order to explore the impact of the super cyclone SIDR for comprehending the ocean ecosystem and health, this study will look at the variation of Chlorophyll-a (Chl-a), Diffuse Attenuation Coefficient (KD), and sea surface temperature (SST) before, during, and after the super cyclone SIDR in the BoB. utilizing MODIS-Aqua datasets, the investigation was conducted utilizing ocean color datasets.

1.1 Objectives of the study

- 1) Analysis chlorophyll concentration of the pre-,during and post-month of the “Sidr” cyclone using Satellite Aqua-MODIS data over the Bay of Bengal basin.
- 2) To observe the diffuse attenuation coefficient (Kd) variability in the Bay of Bengal with respect to Sidr cyclone
- 3) As well as the sea surface temperature (SST) variability in the Bay of Bengal with respect to the cyclone.
- 4) pre-, during and post- “Sidr” months SST and Chlorophyll concentrations variability assessment.
- 5) Variability assessment of Chlorophyll concentrations variability during pre-, during and post“Sidr” months with respect to the diffusive attenuation coefficient (Kd).

2. MATERIALS AND METHODS

2.1. Study Area

SIDR developed in the central Bay of Bengal, southeast of the Andaman Islands, in close proximity to the Nicobar Islands from November 9 and gradually intensified to the shore of Bangladesh at 15 November. So, our conducted study area is the entire Bay of Bengal where the affects of cyclone on chlorophyll concentration is analysed with two of the physical parameter of water (SST, Kd). BoB is a semi enclosed basin in the north Indian Ocean and is forced by seasonally reversing monsoons (S. P. Kumar et al,1996). Its major distinctive feature, however, is the tremendous freshwater influx it receives from rivers. The fresh water influx together with the monsoon winds exert a strong influence on the surface water circulation and stratification. (Sarangi et al., 2015)

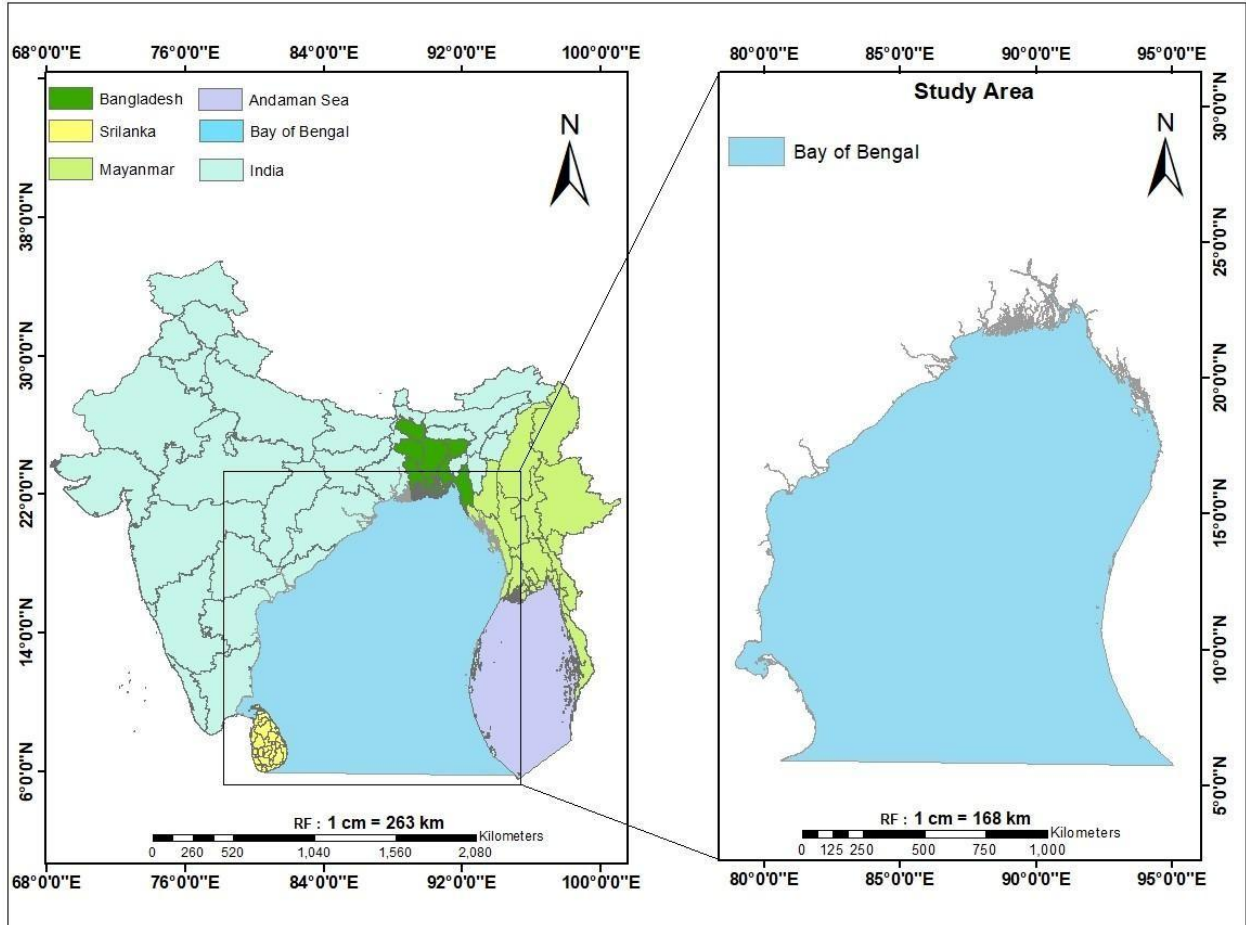


Figure 1: Geographical location of the study area (left, Part of South East Asia; right, Bay of Bengal)

2.2. Data Collection

Conducting in-situ observations during cyclone is very challenging. Therefore, in this work, we are using satellite data to examine the distribution of chlorophyll during the cyclone SIDR. The IBTrACS (International Best Track for Climate Stewardship) of NASA was used in the present study to get cyclone track point data with latitude, longitude and wind speed. On November 9, 10,

11, 12, 13, 14, 15, 16 and 17, 2007, the Quikscat Scatterometer was used to achieve daily wind data that were used to examine the air-sea interaction throughout the SIDR period. The National Aeronautics and Space Administration (NASA) ocean colour web([NASA Ocean Color](#)) was used to acquire composite maps of level-4,monthly data of Chlorophyll(Chl_a) ,Sea Surface Temperature (SST) and Diffused Attenuation Co-efficient(Kd) from satellite MODIS aqua. In this study ,the pre-cyclone period for SIDR was taken to be the month of October of 2007 and the post-cyclone period to be the month of December in the same year. The cyclone occurred in 2007 in the month of November. In this study, satellite image processing software called SNAP-which is primarily used to interpret ocean color data of various levels-was utilized to analyze Kd,Chl_a and SST data. Using the well known GIS program Arc MAP 10.8 , composite images of SST,Kd and Chl_a were shown. Suitable 25 tracking points have been drawn along the composite pre-, during and postcyclone Chl_a, SST and Kd images within the co-ordinates (latitude 5°N-22°N and longitude 80°E and 100°E) ([Banglapedia](#)) .

2.3. Statistical Analysis

From the study area 25 stations were randomly selected from nearly onshore to the offshore region. These stations sea surface temperature, diffuse attenuation coefficient and chlorophyll concentration data were used for Regression analysis method. As SST and Kd are two of the water parameters that had variation due to the cyclone storm. So, univariate regression focuses here on determining relationship between Chlorophyll concentration and SST as well as Chlorophyll concentration and Kd of October(pre-storm), November (during-storm) and December (Post-storm).

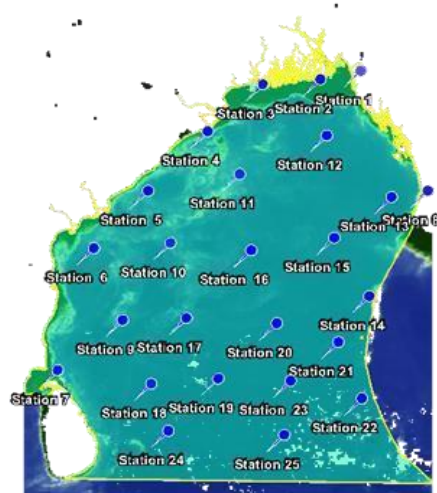


Figure 2: Elected tracking point of 25 stations for each month Chl_a,SST and Kd

3. RESULT AND DISCUSSION

3.1. Formation and Path of Tropical Cyclone Sidr

November 9 - An air mass disturbance with weak low-level circulation began developing in the central Bay of Bengal, southeast of the Andaman Islands, in close proximity to the Nicobar Islands. Moderate upper-level wind shear inhibited its organized development, however strong diffluence aided in developing convection.

November 11- The anomalous weather system was still somewhat south of the Andaman Islands. A better defined cyclonic circulation developed when the vertical shear began decreasing. Based on that, a ‘tropical cyclone formation alert’ was issued for the region. Later that day, when winds reached 76-107 knots , the India Meteorological Department (IMD) designated the system as a tropical depression. The system intensified as it moved slowly in a north-westward direction.

November 12 - As the system further intensified, IMD further upgraded the weather system to a ‘severe cyclonic storm’ and named it ‘Sidr’ (‘hole’ or ‘eye’ in Sinhalese language).

November 15 - By morning, Cyclone Sidr had moved considerably northward towards India’s eastern border with Bangladesh. It had strengthened to reach with peak sustained winds of 108140 knots. According to JTWC best track, Cyclone Sidr subsequently reached peak wind velocities of up to 260 km/h. Later that day, it appeared that the brunt of the cyclone’s force would be felt by the less populated areas of the Sundarbans mangrove forests, which stretch along the western coast of Bangladesh. However, that did not happen. Around 11PM, with sustained winds of 215 km/h, Sidr made direct landfall in the district of Bagerhat, a highly-populated area of the country. A catastrophic storm surge flooded the area and caused most of the deaths and damage.

November 16 – Cyclone Sidr weakened considerably as it moved over land. (drgeorgepc.com)

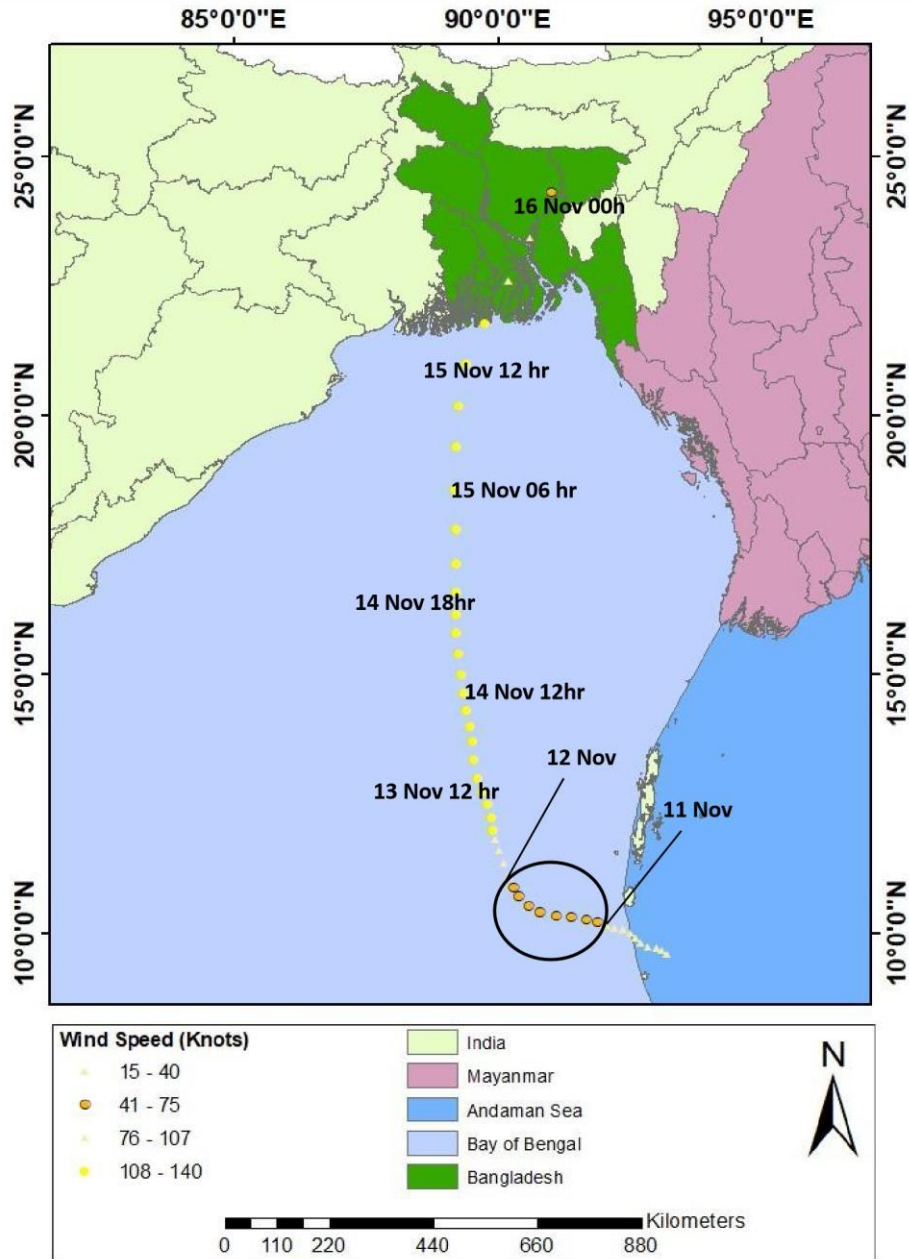


Figure 3: The study region and the tracks of tropical cyclones occurring 9th November – 17th November.

Sidr was classified as Category - V storm (as per Saffir-Simpson Hurricane Scale) on November 15, 2007, with wind speeds up to 135 knots (250 kmph). (Unisys Weather). The Map showing the track of Cyclone Sidr and mooring locations. The study region and the tracks of tropical cyclones occurring in 2007 from 9th November to 16 November with the Maximum windspeed 108-140 knot. 15-40 knots was the minimum windspeed that made landfall in Bangladesh.

Table1: Spatial Windspeed data of Cyclone Sidr track. (NR= Not Reported, TS= Tropical Cyclone)

Nature	latitude	Longitude	Distance to Land (Km)	Wind Speed (Knots)
NR	9.6	93.2	499	15
NR	9.65249	93.1125	514	17
NR	9.7	93	519	20
NR	9.74251	92.85	524	22
TS	9.8	92.7	545	25
TS	9.89998	92.6	560	27
TS	10	92.5	575	30
TS	10.0575	92.3575	591	32
TS	10.1	92.2	602	35
TS	10.12	92.052	609	40
TS	10.1333	91.9333	621	45
TS	10.1429	91.8129	628	47
TS	10.4	91.45	674	50
TS	10.4175	91.2962	688	55
TS	10.45	91.15	687	60
TS	10.5137	90.785	710	65
TS	10.85	90.45	692	70
TS	10.9425	90.3925	698	75

TS	11.3	90.1	685	80
TS	11.425	90.0463	684	92
TS	11.55	90	668	105
TS	11.65	89.9712	668	107
TS	12	89.7	660	110
TS	12.1212	89.6787	652	112
TS	12.25	89.65	638	115
TS	12.8675	89.6	604	115
TS	13	89.55	598	115
TS	13.1712	89.5212	595	115
TS	13.6	89.5	572	115
TS	14	89.475	552	115
TS	14.4	89.45	535	115
TS	14.5675	89.425	538	117
TS	15	89.4	526	120
TS	15.4463	89.3713	518	125
TS	15.9	89.1	540	130
TS	16.085	89.0962	521	135
TS	16.55	89.1	475	140
TS	17.07	89.1	431	140

TS	17.65	89.1	383	140
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Data Source: Source: International Best Track Archive for Climate Stewardship (IBTrACS)

3.2. Air Sea Interaction During the Storm Surge

The wind's direction and speed have provided clear explanations of the cyclone's eye, intensity, and migration path. Images show that the storm originated in the southernmost part of the Bay of Bengal and moved toward the Bangladeshi shore. From the scatterometer's collected wind speed images, similar information has been deduced. The highest wind speed was measured at about 140 knots (1 Knot = 1.85 km/hr). With the use of wind speed data, it has been thoroughly understood how the chlorophyll patch is forced from the offshore to the coast.

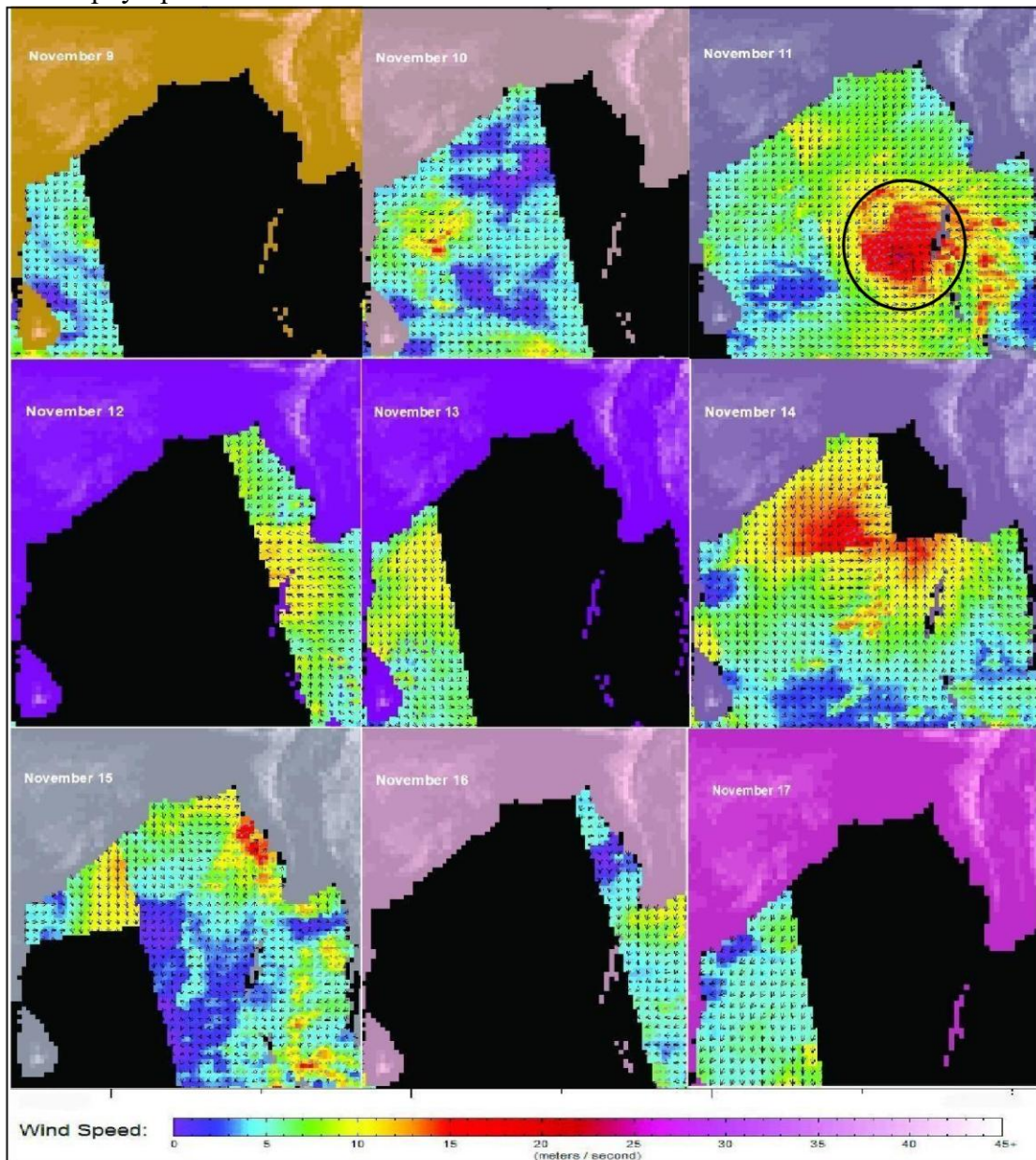


Figure 4: 'Quickcat scatterometer' derived wind speed and wind vector maps during 9 November, 2007 to 17 November, 2007 over

The 11th November windspeed was between 15-20 m/s near the southeast of the Andaman Islands with a weak low-level circulation near the Nicobar Island (figure 4). After 3 days, the high pressured wind striking forward to the onshore began to intensify quickly as moves northwestward.

3.3. Parameters Variability in response to Sidr 3.3.1. Chlorophyll Response

Composite image of Chl-a shows low concentrations in most of the Bay during the pre-cyclone period around 0.04-0.91 mg/m³. It was higher near the coastal belt about 2.94-6.43mg/m³. During cyclone the concentration increased dominantly along the the coast of the bay due to low SST, high wind and vertical mixing ranging from 3.49-5.69mg/m³ and a increased level of Chl_a is noticed along the cyclone track .Along the central coast highest concentration of chlorophyll accumulation is seen ranging at around 37mg/m³. After the cyclone the concentration started to decrease along the coast. An average concentration of 2.70-4.69mg/m³ Chl_a is seen along the central coast of the Bay.

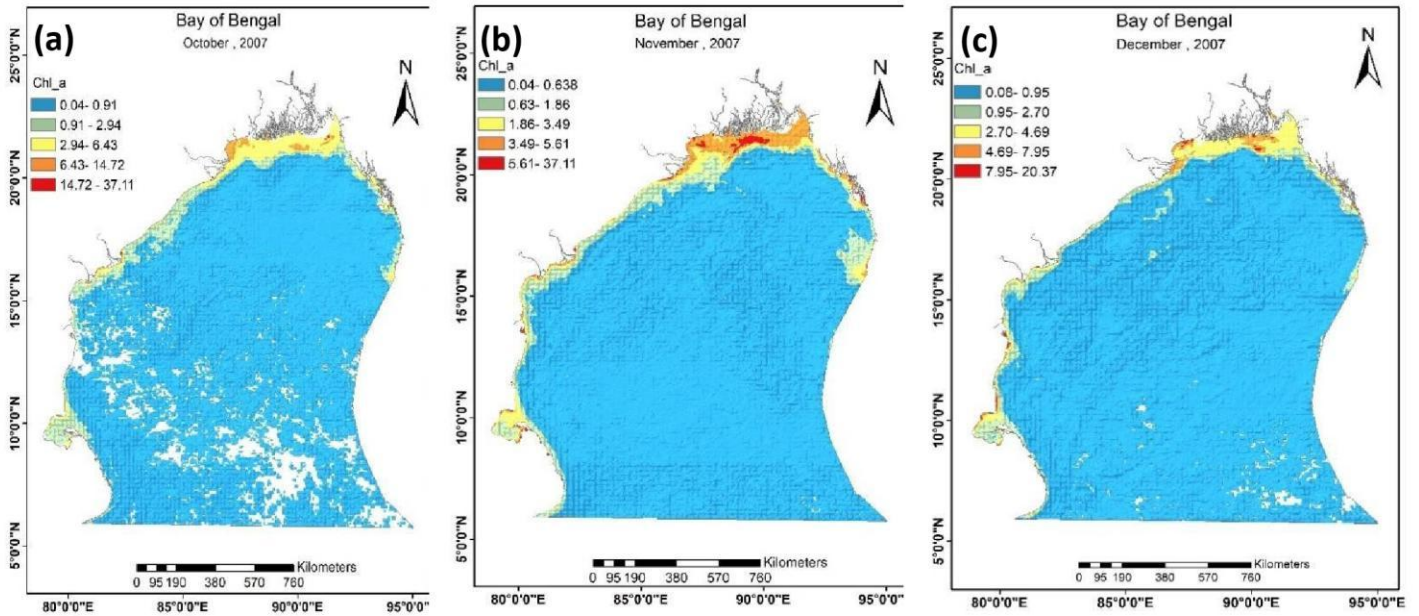


Figure 5: Composite image of Chl_a (a) pre-month; October, (b) affected month; November, (c) post-month; December

3.3.2. SST response

Sea Surface Temperature composite image shows very high temperature during pre-cyclone period and very low during post-cyclone period . During pre-cyclone period the north-east and south-east part of the bay showed highest temperature ranging from 29.33-31.11°C . In the central bay it reduces a little to 28-29°C. Temperature decreases as we move further offshore. During cyclone the temperature decreases along with the cyclone track .During this time the coastal region has the

lowest temperature of 21-27°C and in the north-eastern part it ranges from 27.48-28.11°C. During the post cyclone period SST image shows a gradual increase in temperature from coast to offshore. The coastal region presents lowest temperature while the open ocean shows higher temperature. So, SST decreases in the post-cyclone with reference to the pre-cyclone period.

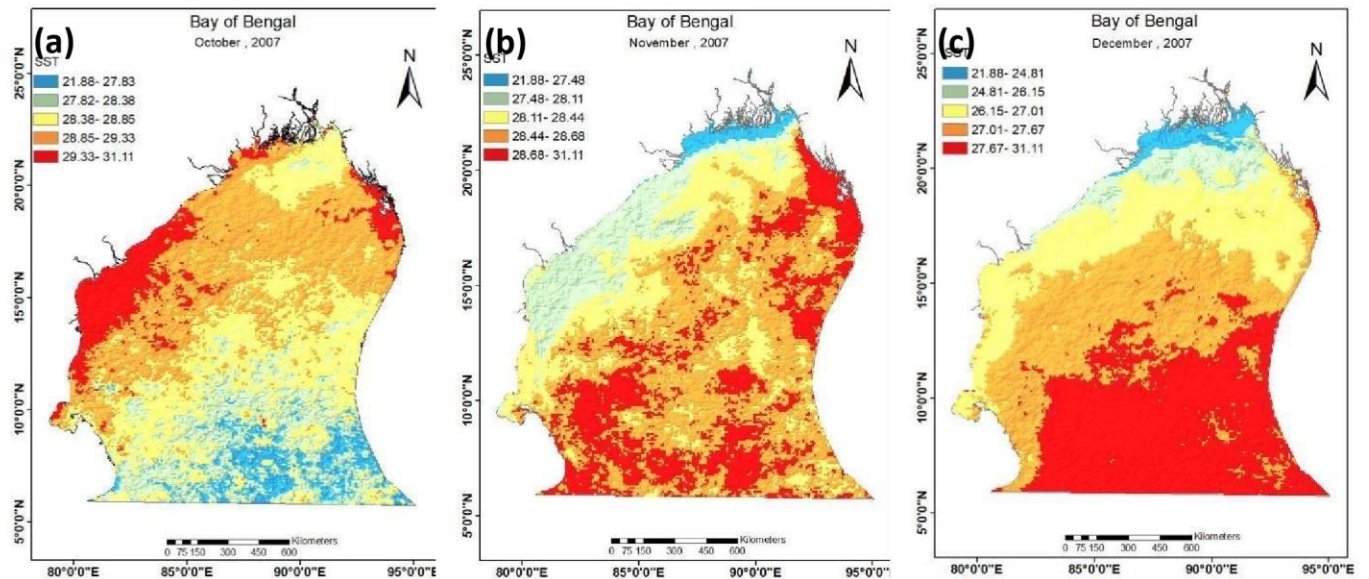


Figure 6: Composite image of SST (a) pre-month; October, (b) affected month; November, (c) post-month; December

3.3.3. Kd Response

The diffuse attenuation coefficient (K_d) is a measure of how light dissipates with depth in water. The K_d is an indicator of the turbidity of the water column, and is directly related to the concentration of scattering particles into the water column (noaa.gov). There is a linear relationship between attenuation coefficient and turbidity, that an increase in water turbidity leads to an increase in Diffused Attenuation Coefficient (K_d). From the composite maps of K_d the change in turbidity is analysed. During pre-cyclone period the K_d is much less all over the Bay. Near coastal area it increases to 0.11-0.34 m^{-1} . During cyclone SIDR there is a high increase in K_d in coastal area and only the central part of the central coast has the highest ranging K_d of 2.20-4.15 m^{-1} . After the storm in December there is a small decrease in K_d in the coastal zone. For the rest of the Bay, K_d remains much unchanged ranging from 0.01-0.11 m^{-1} . This leads us to believe turbidity increases during cyclone from coast to off-shore.

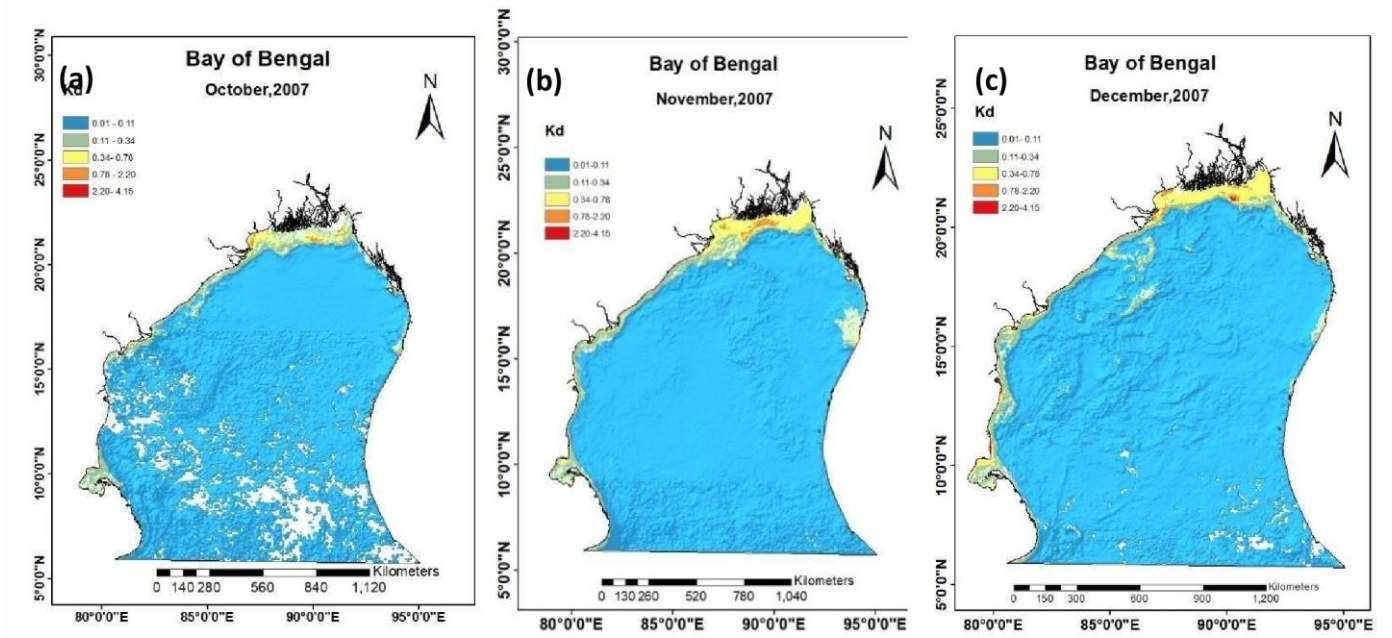


Figure 7: Composite image of Kd (a)pre-month; October, (b) affected-month; November, (c)post-month; December.

3.4. Chlorophyll concentrations correlation with cyclone influencing variable

The main aim of regression analysis is to visualize the underlying relationship between water parameters that adjoint with cyclone and chlorophyll concentrations. Univariate linear regression focuses on determining the relationship between one explanatory variable and one dependent variable. Here the water parameters (SST, Kd) are regarded as control variable over the response variable Chlorophyll.

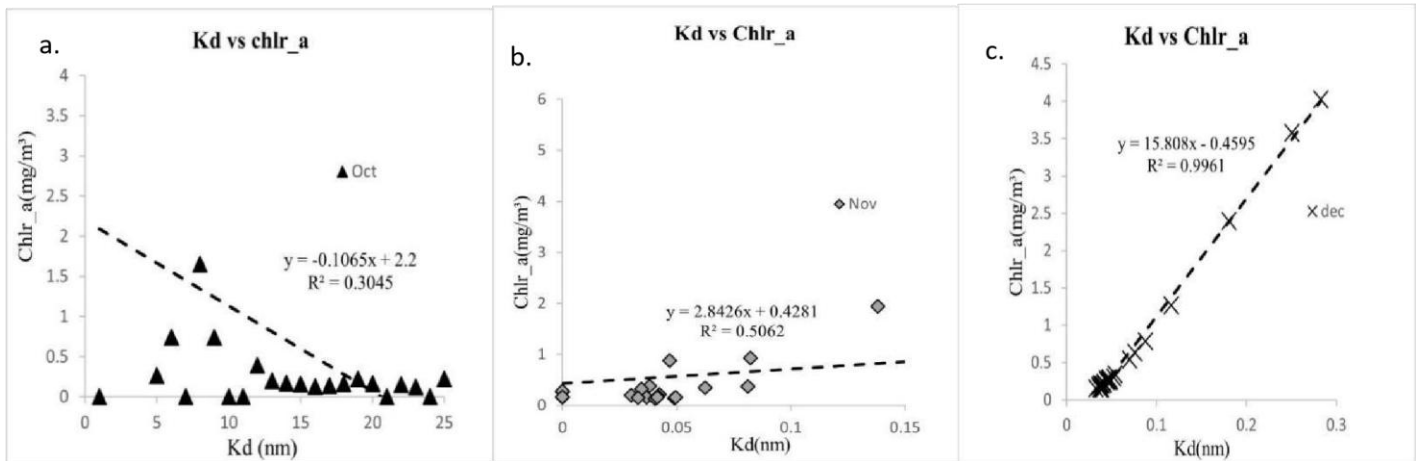


Figure 8: For Kd-Chlr_a The co-efficient of determination (R^2) increases gradually from 0.3045 to 0.9961 due to the cyclone that is close to 1 (a,b,c)

The linear regression model: $Y_i = bX_i + C$, where, X_i = Control Variable of the stations; Y_i = Response variable that varies as X_i does. C is the random component of the regression handling the residue.

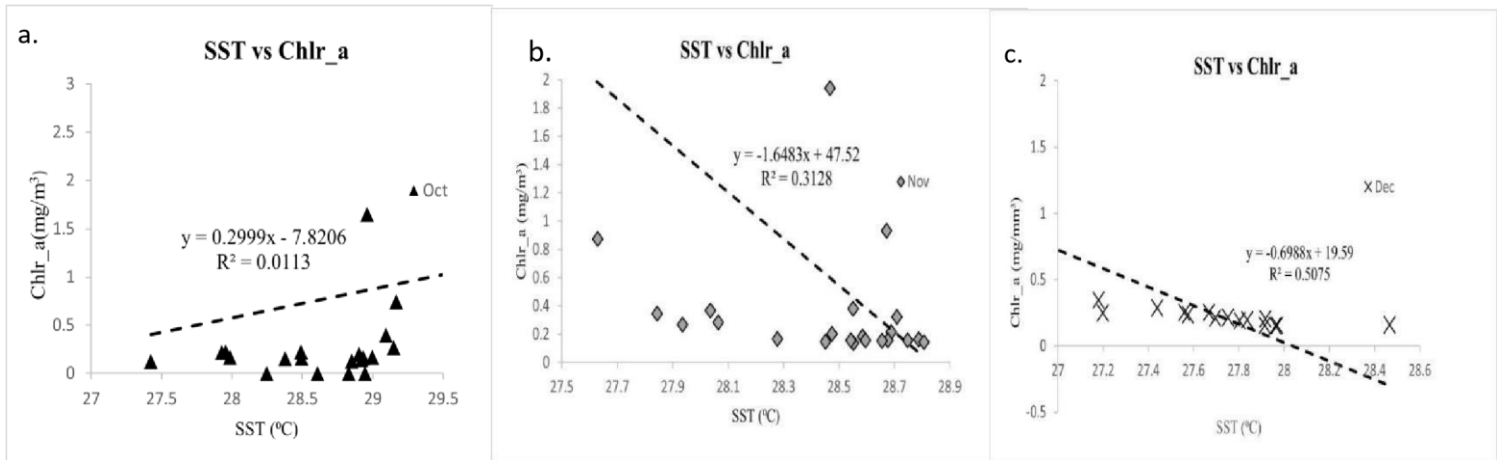


Figure 9: The R^2 value (co-efficient of determination) range is 0.0113 to 0.5075 for SST input variable on Chl_a. (a,b,c)

Fig a,b In October the diffuse attenuation coefficient explain chlorophyll concentration only 30%. The rate for SST is only 1%. But these stations next month coefficient value is 0.5062 (fig c) for Kd vs Chl_a and 0.3128 (fig d) for SST vs Chl_a. That means the increasing temperature due to cyclone formation had an influence on the chlorophyll distribution. It's also clear that during and after the storm the control of light penetration rate with depth decreased from 20-25nm to 0.5 to 0.1nm. The upmost Kd and chlorophyll concentrations coefficient is 0.9961 during post cyclone month. Where pre-cyclone period chlorophyll concentrations change with SST is 0.0113, in post cyclone month this coefficient rose to 0.5078.

4. CONCLUSION

Despite numerous cyclones passing over the area in 2007, cyclone Sidr had a significant impact on the chl a concentration in the waters of the Bay of Bengal. This single-track, slowly moving storm lingered in the Bay of Bengal for more than five days. Strong upwelling was created by the cyclone's prolonged forcing time, which also significantly altered the distribution of chlorophyll. During cyclonic occurrence, a rise in the content of chlorophyll up to 2.5 mg/m³ has been seen. In the coastal seas, there has been movement and accumulation of intense concentrations of chlorophyll. A reduction of up to 21°C in sea surface temperature has been noted. These post cyclone regions would be significant potential fishing zones (PFZs) and can be well predicted for the fisher-folk community of our nation and others. Wind speed images are seen and well correlated with the movement and direction of high chlorophyll patches, which acted as a triggering force for chlorophyll enhancement and induces the productivity of the coastal water.

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